

Ayush Shinde

MACHINE LEARNING ENGINEER | AI & MLOps | DATA SCIENTIST | DATA ENGINEER |

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SUMMARY

ML Engineer with 2+ years of experience developing **machine learning models** and **data pipelines** for **fraud detection** and **predictive maintenance**. Skilled in **Python, SQL, PySpark, Scikit-learn, XGBoost, LSTM, Databricks, AWS, and Azure**. Experienced in **feature engineering**, **data processing**, **model deployment**, and **CI/CD** for ML applications. Worked with **fraud analysts**, **data engineers**, and **model-risk teams** to **validate models**, **investigate results**, and **support model releases**.

SKILLS

Languages: Python, SQL, PySpark, Scala, Bash

ML & AI Frameworks: Scikit-learn, XGBoost, TensorFlow, PyTorch, Pandas, NumPy, SciPy

Big Data & Data Engineering: Apache Spark, Databricks, AWS EMR, Hadoop, ETL Pipeline Design, Data Validation/Quality Framework

Cloud & Infrastructure: AWS (S3, SageMaker, EMR, EC2), Azure (AKS), Docker, Kubernetes, Microservices Architecture

MLOps & DevOps: CI/CD (GitHub Actions), Docker, Model Lifecycle Management, Model Validation, Infrastructure as Code (IaC)

Databases & Analytics: PostgreSQL, AWS Redshift, Snowflake, Tableau, Data Visualization

Domains & Core Concepts: Fraud & Anomaly Detection, Predictive Maintenance, Time-Series Analysis (LSTM), Distributed Computing, Real-Time Inference Pipelines, Feature Engineering

EXPERIENCE

JPMorgan Chase & Co., TX | ML Engineer | Feb 2026 – Current

- Engineered **fraud classification models** using **Scikit-learn**, **building transaction, customer, merchant**, and **spending-pattern** features on curated datasets that improved identification of **high-risk transactions** during **model validation**.
- Built **data preparation pipelines** using **SQL, PySpark**, and **Databricks** to combine transaction, customer, merchant, and fraud-alert data, adding **validation checks** that reduced recurring **data-quality issues** by 25%.
- Reduced batch processing time from 48 to 38 minutes by profiling **Python** and **Spark transformations**, eliminating redundant calculations, and tuning **data-processing steps**.
- Collaborated with fraud analysts, data engineers, and model-risk teams to review **false-positive cases**, compare **model results**, and document changes, helping prioritize higher-risk alerts for manual investigation and supporting controlled **model releases**.

Siemens, India | ML Engineer | Aug 2022 - Jul 2024

- Developed **predictive maintenance models** using **LSTM** and **XGBoost**, achieving 94% fault detection accuracy across 5 manufacturing plants and reducing unexpected equipment downtime by 27%.
- Engineered **PySpark pipelines** processing 500GB+ of **IoT telemetry** from 10,000+ edge devices, reducing data preparation time by 60% for downstream **ML workloads**.
- Built **containerized inference APIs** with **Docker** and **Azure AKS**, enabling **real-time anomaly detection** for 2,000+ production assets while reducing inference response time by 40%.
- Migrated on-premise **ML infrastructure** to **AWS**, including **S3, EMR**, and **SageMaker**, lowering annual infrastructure expenses by ₹35 Lakhs and improving deployment scalability.
- Created **Tableau dashboards** used by 50+ plant managers, improving maintenance planning efficiency and contributing to annual cost savings of approximately ₹1.2 Crore.
- Implemented **CI/CD pipelines** using **GitHub Actions**, reducing deployment errors by 50% and shortening model release cycles by 40%.

PROJECTS

RAG Support Insights & LLM Evaluation | (Python, LangChain, OpenAI API, FAISS, Streamlit, FastAPI)

- Constructed a **RAG pipeline** using **LangChain** and **GPT-3.5-turbo** on 28K+ support records, embedding documents into **FAISS** for **issue classification** and **next-best-action recommendations**.
- Evaluated 200 test queries with a **RAGAS-inspired relevance and faithfulness framework**, reaching 83% relevance while cutting hallucinations by 31% after deployment.

AI Logistics Operations Agent | (Python, OpenAI API, FastAPI, Streamlit, Docker)

- Designed an **AI operations agent** that interpreted **natural-language requests** and generated **structured recommendations** for shipment delays and fulfillment exceptions.
- Packaged the **FastAPI backend** and **Streamlit interface** in **Docker**, adding **validation** and **error handling** to support repeatable testing and demos.

EDUCATION

Master of Science in Data Science | University of Texas at Arlington, Arlington, TX | **May 2026**

B.E. – Electronics & Telecommunication Engineering | AI & ML Honors | University of Mumbai | **Aug 2024**

CERTIFICATIONS

Certified Associate Machine Learning Engineer – AWS | **Databricks Fundamentals – Databricks**

Advanced Google Data Analytics – Google